

GRADISTE, Croatia

WMO: 143820

Lat: 45.1592N

Lon: 18.7036E

Elev: 322

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.3	18.6	7.8	8.3	15.6	12.8	10.7	20.7	18.8	37.9	16.5	36.6	4.3	70	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	93.7	71.2	90.6	70.8	87.6	69.7	74.1	87.6	72.6	86.2	71.2	84.3	4.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	111.1	79.9	68.3	105.1	78.7	66.8	99.8	77.0	37.8	88.2	36.4	86.3	35.2	84.9	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	13.8	11.7	6.6	97.6	8.2	4.2	0.7	100.6	-4.2	103.0	-8.8	105.3	-14.8	108.4		
			6.0	76.8	8.2	1.4	0.1	77.8	-4.7	78.7	-9.3	79.4	-15.2	80.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	34.4	37.1	46.0	55.7	63.1	70.1	73.1	72.9	64.1	55.0	46.4	36.8
	DBStd	15.63	9.17	9.54	8.77	6.77	6.74	6.49	5.64	5.64	6.74	7.51	8.08	7.94
	HDD50	1677	485	368	180	26	3	0	0	0	38	161	416	
	HDD65	4605	947	782	591	284	116	27	6	96	317	558	875	
	CDD50	3378	3	6	55	198	408	603	717	711	423	194	54	6
	CDD65	828	0	0	0	6	56	179	258	252	69	8	0	0
	CDH74	8637	0	0	3	118	596	1790	2683	2672	687	87	1	0
	CDH80	3417	0	0	0	20	145	665	1155	1212	214	6	0	0
Wind	WSAvg	4.9	5.0	5.5	6.1	5.4	5.3	4.7	4.7	4.3	4.5	4.3	4.5	4.9
Precipitation	PrecAvg	29.3	1.9	1.8	1.9	2.4	3.0	3.3	3.0	2.5	2.7	2.3	2.4	2.2
	PrecMax	39.5	3.5	3.2	3.3	6.4	7.0	9.3	8.3	7.2	8.1	5.8	4.6	5.4
	PrecMin	14.3	0.4	0.2	0.3	0.4	1.1	1.3	0.6	0.1	0.0	0.0	0.2	0.1
	PrecStd	6.8	0.8	0.8	0.9	1.4	1.6	1.8	1.9	1.8	1.9	1.6	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.5	64.6	73.3	83.3	88.4	93.5	98.0	99.0	91.1	80.5	71.5	61.4
		MCWB	51.1	51.4	55.1	62.2	66.4	72.4	71.6	70.9	68.1	63.5	58.9	52.9
	2%	DB	54.3	58.2	69.0	77.2	84.0	90.3	93.6	94.3	85.8	76.3	66.8	56.4
		MCWB	47.7	48.9	53.3	58.7	66.1	71.5	71.5	70.9	66.2	62.0	56.6	49.9
	5%	DB	50.2	54.1	64.9	73.3	80.5	87.0	90.2	90.5	81.1	72.3	62.3	51.7
		MCWB	45.5	46.7	51.8	57.2	64.5	70.4	71.5	70.2	64.9	60.6	54.5	46.7
	10%	DB	46.1	49.7	60.4	69.4	76.9	83.8	86.8	87.0	77.4	68.0	58.4	47.4
		MCWB	42.5	44.6	49.7	55.3	62.4	68.7	70.1	69.1	63.8	58.6	52.6	43.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	52.3	56.7	62.9	69.3	75.3	75.9	75.5	70.2	65.7	59.8	53.8
		MCDB	58.7	61.9	69.8	80.9	81.9	87.8	89.4	87.9	85.1	76.9	69.2	60.0
	2%	WB	48.4	49.8	54.4	60.3	67.4	73.3	74.2	73.5	68.2	63.3	57.4	50.2
		MCDB	53.6	57.4	65.4	73.9	80.0	86.0	88.0	88.1	81.4	73.1	64.3	55.4
	5%	WB	45.5	47.3	52.7	58.4	65.6	71.7	72.6	71.9	66.5	61.4	55.5	47.0
		MCDB	49.5	52.6	62.8	70.3	78.0	84.2	86.4	86.0	77.5	70.1	62.1	51.3
	10%	WB	42.9	45.0	50.6	56.6	63.8	70.2	71.2	70.5	64.8	59.7	52.8	44.2
		MCDB	45.6	49.1	59.6	66.8	75.0	81.5	84.5	83.5	74.7	66.9	57.9	47.2
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.9	17.0	19.8	19.7	19.8	21.3	22.5	19.9	18.5	14.4	10.0
		MCDBR	17.6	20.8	25.3	27.8	25.5	24.9	27.2	29.1	27.0	25.5	20.9	17.3
		MCWBR	12.1	12.9	13.1	12.6	10.5	9.3	9.3	10.1	10.4	12.9	12.4	11.8
	5% WB	MCDBR	17.0	18.9	22.5	23.9	23.5	23.0	24.6	25.2	23.3	23.3	19.2	16.2
		MCWBR	11.8	12.1	12.4	11.3	10.3	9.3	9.3	9.9	9.6	12.3	11.8	11.6
Clear-Sky Solar Irradiance	taub	0.341	0.369	0.401	0.458	0.445	0.442	0.445	0.448	0.431	0.397	0.360	0.341	
	taud	2.356	2.272	2.221	2.073	2.169	2.218	2.206	2.203	2.217	2.307	2.364	2.363	
	Ebn at Noon	239	253	261	254	263	264	261	255	248	239	229	225	
	Edh at Noon	25	32	39	49	46	44	44	43	39	31	25	23	
All-Sky Solar Radiation	RadAvg	396	642	1036	1428	1748	1925	1958	1747	1211	834	490	322	
	RadStd	60	89	130	145	154	164	148	178	138	90	46	50	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	+1.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+1.21	N/A	N/A	N/A	+0.74	+0.77	-216	-329	+207	+110

Nomenclature: See separate page